STAT 526 HW 9

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Problem 0

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Problem 1

To assess the gender risk of death from coronary heart disease (CHD), a study was performed controlling for genetic and environmental factors by using fraternal twins, one male and one female. The age at which a male/female twin died of CHD was recorded. The dataset twin contains the following samples from this study.

a.)

Calculate the Kaplan-Meier estimates of the survival functions for the male group.

Our estimate function is:

$$\hat{S}(t) = \prod_{i:t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

We can make a table for that:

Age (t_i)	At Risk (n_i)	Deaths (d_i)	Censored	$1 - d_i/n_i$	$\hat{S}(t)$
0	12	0	0	-	1.000
49	12	1	0	0.917	0.917
50	11	0	1	1.000	0.917
56	10	1	0	0.900	0.825
61	9	0	1	1.000	0.825
67	8	0	1	1.000	0.825
68	7	0	1	1.000	0.825
69	6	1	0	0.833	0.688
70	5	1	0	0.800	0.550
74	4	2	0	0.500	0.275
75	2	1	0	0.500	0.138
81	1	1	0	0.000	0.000

b.)

> Using the output below, state the hypothesis that gender is a risk factor of death from CHD, and test the hypothesis while controlling for genetic and environmental factors.

This is Log-Rank Test of homogeneity.

The null hypothesis is that there is no differences between the survival probability between the male group and female group.

The alternative hypothesis there is a significant difference between the survival probability between the male group and female group.

We can do the test. From the output, we observe 7 female deaths and 8 male deaths with expected deaths to be 7.91 for female and 7.09 for male.

We need to calculate the test statistic. We first solve for variance given our output:

$$\frac{(7-7.91)^2}{V} = 0.299 \implies v = 2.76957$$

We can calculate the test statistic $((7-7.91)^2 + (8-7.09)^2)/\sqrt(2.76957)$ or simply use the R output of 0.299 on 1 degree of freedom.

```r
pchisq(0.299, 1, lower.tail = FALSE)
```

```
## [1] 0.58451
```

For $\chi^2 = 0.299$ with 1 df, the p-value is approximately 0.5845.

Since p-value is greater than confidence level 0.05, we fail to reject the null hypothesis. We believe that there is no significant evidence that the two groups have different survival probabilities.

c.)

> An accelerated failure time model is fitted to the data as shown below. Using the output below, predict the probability for a female with CHD to survive past 75.

Survival function is

$$S(t) = \exp\left(-t \cdot \lambda\right) = \exp\left(-t \cdot e^{-\mu}\right) = \exp(-t \cdot e^{-(\beta_0 + \beta_1 x)})$$

Female is baseline so $\mu = \text{Intercept} = 4.772$.

The scale parameter is $\lambda = e^{-\mu} = e^{-4.772} \approx 0.008463$. Then:

$$S(75) = \exp(-75 \cdot 0.008463) = \exp(-0.634725) \approx 0.530$$

The predicted survival probability to survive past age 75 is around 53%.

d.)

> A proportional hazards model is fitted to the data as shown below. For a twin consisting of a male and female, what is the effect of gender on the survival?

We see outputs coef (β) = 0.26 and exp(coef) (HR) = 1.30. The exp(coef) is the hazard ratio meaning we have 30% higher risk of death from CHD as a male compared to female.

However, the p-value is 0.62 and our confidence interval includes 1, meaning we do not have significant evidence of the effect of gender on survival.

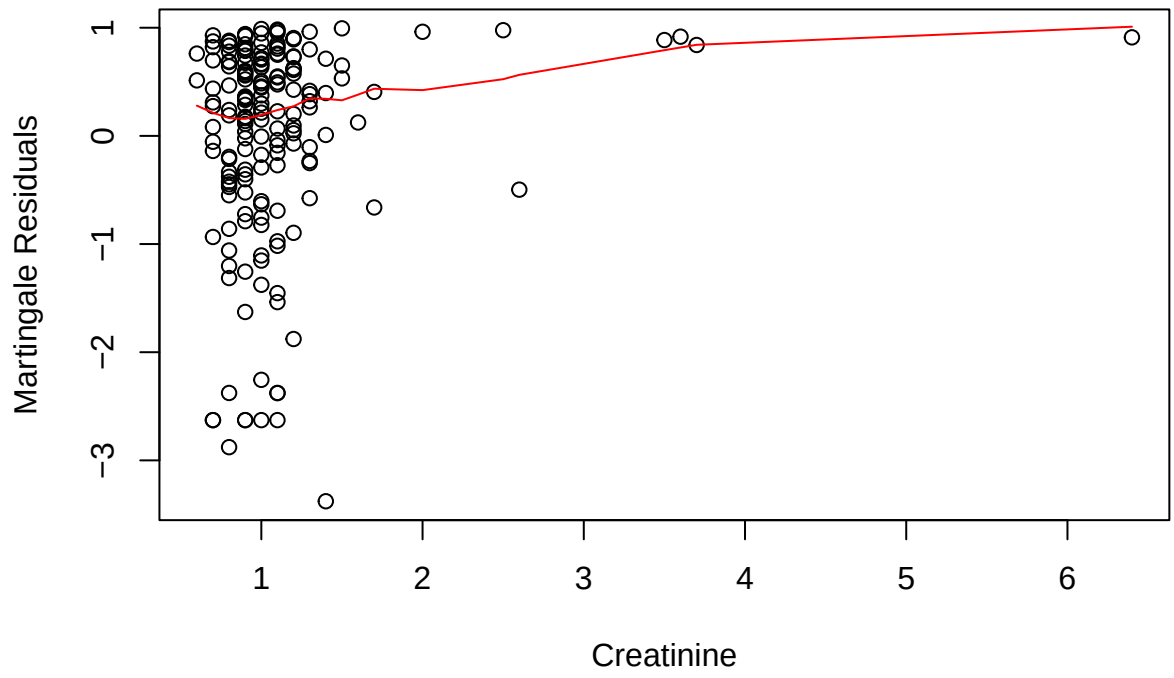
Problem 2

Consider the dataset mgus in the library survival (use ?mgus for information on the dataset). Your goal is to model the survival time of these patients using Cox proportional hazard model (ignore variables pcdx and pctime, which have a large proportion of missing data). Verify the plausibility of a proportional hazard model for this dataset. Select important predictors in the model, and their functional form. Evaluate the quality of fit (optional). Interpret model parameters and discuss the results. Hint: i) logarithm transformation of creat is suggested due to its high skewness ii) please use resid() function to obtain martingale residuals. Use visual diagnosis of martingale residuals (under null model) vs predictor. iii) To diagnose proportional hazard assumption with respect to continuous predictor, you can convert this continuous predictor to categorical variable (say partition based on empirical quantity), then use the plot of log(-logS(t)) vs log(t).

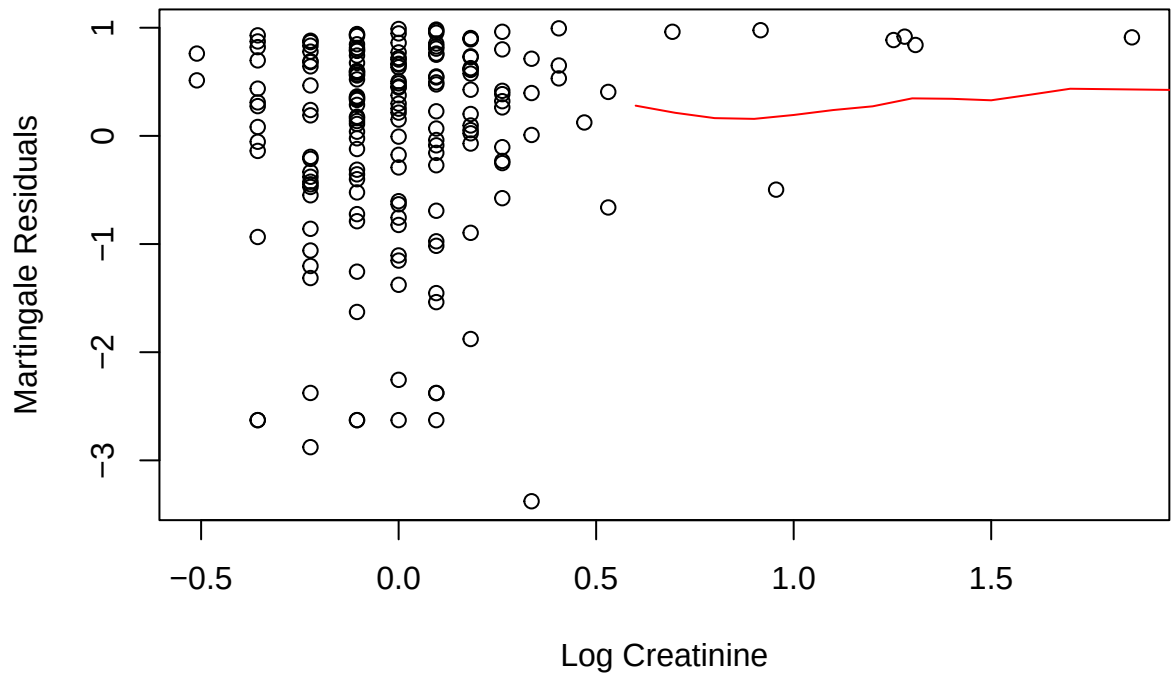
```r
library(survival)
library(survminer)
```

```
## Warning: package 'survminer' was built under R version 4.5.3

## Loading required package: ggplot2

## Loading required package: ggpubr

## Warning: package 'ggpubr' was built under R version 4.5.3

##
## Attaching package: 'survminer'

## The following object is masked from 'package:survival':
##
##     myeloma
```

```r
# Ignore pcdx and pctime as per instructions
df <- mgus[, !(names(mgus) %in% c("pcdx", "pctime"))]
df <- na.omit(df) # get rid of NA

# Null model to obtain martingale residuals
null_model <- coxph(Surv(futime, death) ~ 1, data = df)
df$mart_resid <- resid(null_model, type = "martingale")

# Transform creat as suggested by hint (i)
df$log_creat <- log(df$creat)

# Visual diagnosis of martingale residuals (under null model) vs predictor (creatine) by hint (ii)
plot(df$creat, df$mart_resid, xlab = "Creatinine", ylab = "Martingale Residuals")
lines(lowess(df$creat, df$mart_resid), col = "red")
```




```
plot(df$log_creat, df$mart_resid, xlab = "Log Creatinine", ylab = "Martingale Residuals")
lines(lowess(df$creat, df$mart_resid), col = "red")
```



Yeah creat seems important. It doesn't seem super straight, but seems more straight horizontal with log creat.

We now fit the full model:

```
# Full model
fit_full <- coxph(Surv(futime, death) ~ age + sex + hgb + alb + log_creat, data = df)
summary(fit_full)
```

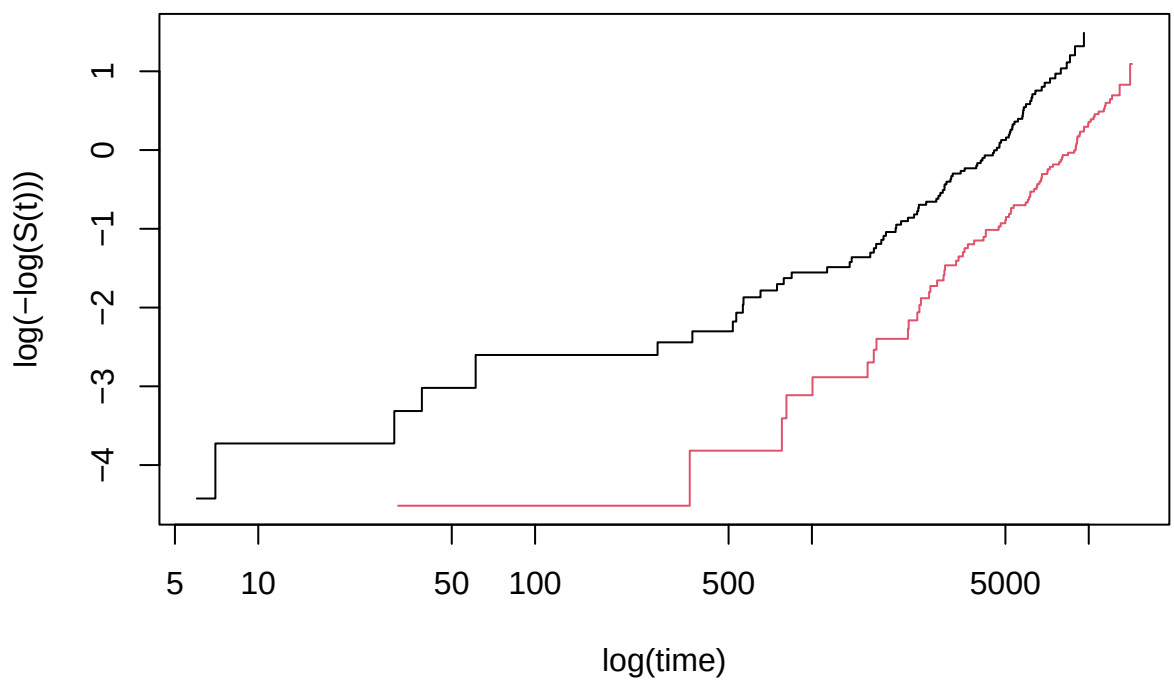
```
## Call:
## coxph(formula = Surv(futime, death) ~ age + sex + hgb + alb +
##     log_creat, data = df)
##
##   n= 176, number of events= 165
##
##                coef exp(coef)  se(coef)      z Pr(>|z|)
## age        0.070298  1.072828  0.008521  8.250   <2e-16 ***
## sexmale    0.176282  1.192774  0.170935  1.031   0.3024
## hgb       -0.114250  0.892035  0.060117 -1.900   0.0574 .
## alb       -0.245500  0.782314  0.201138 -1.221   0.2223
## log_creat  0.539744  1.715567  0.310379  1.739   0.0820 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##           exp(coef) exp(-coef) lower .95 upper .95
## age          1.0728     0.9321    1.0551     1.091
```

```
## sexmale      1.1928     0.8384    0.8532     1.667
## hgb          0.8920     1.1210    0.7929     1.004
## alb          0.7823     1.2783    0.5274     1.160
## log_creat    1.7156     0.5829    0.9337     3.152
##
## Concordance= 0.707  (se = 0.023 )
## Likelihood ratio test= 94.41  on 5 df,   p=<2e-16
## Wald test            = 88.75  on 5 df,   p=<2e-16
## Score (logrank) test = 93.99  on 5 df,   p=<2e-16
```

Concordance is high at 0.707, it is good for fit. We can perform likelihood ratio test here using ANOVA (between null and full model) to evaluate the quality of fit even more.

Age seems to be the only significant one. We can try testing for parallelism (like hint iii says):

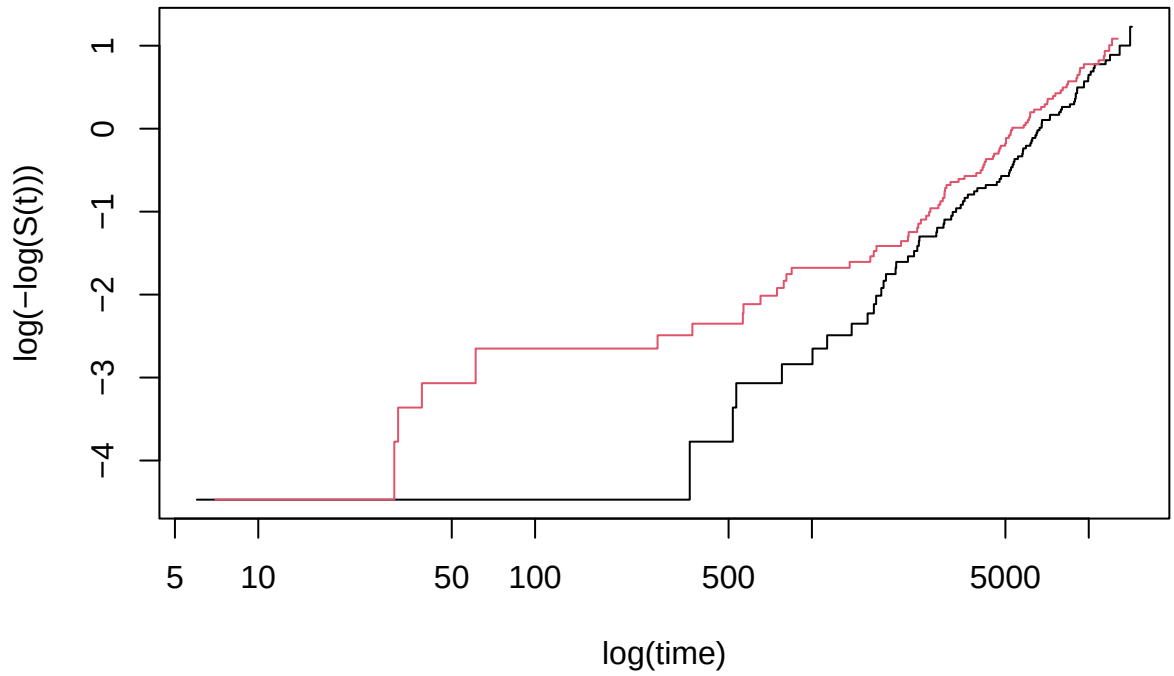
```r
# Partition age into categories
df$age_cat <- ifelse(df$age > median(df$age), "Old", "Young")
surv_fit <- survfit(Surv(futime, death) ~ age_cat, data = df)

# Log-Log Plot
plot(surv_fit, fun = "cloglog", col = 1:2, xlab = "log(time)", ylab = "log(-log(S(t)))")
```



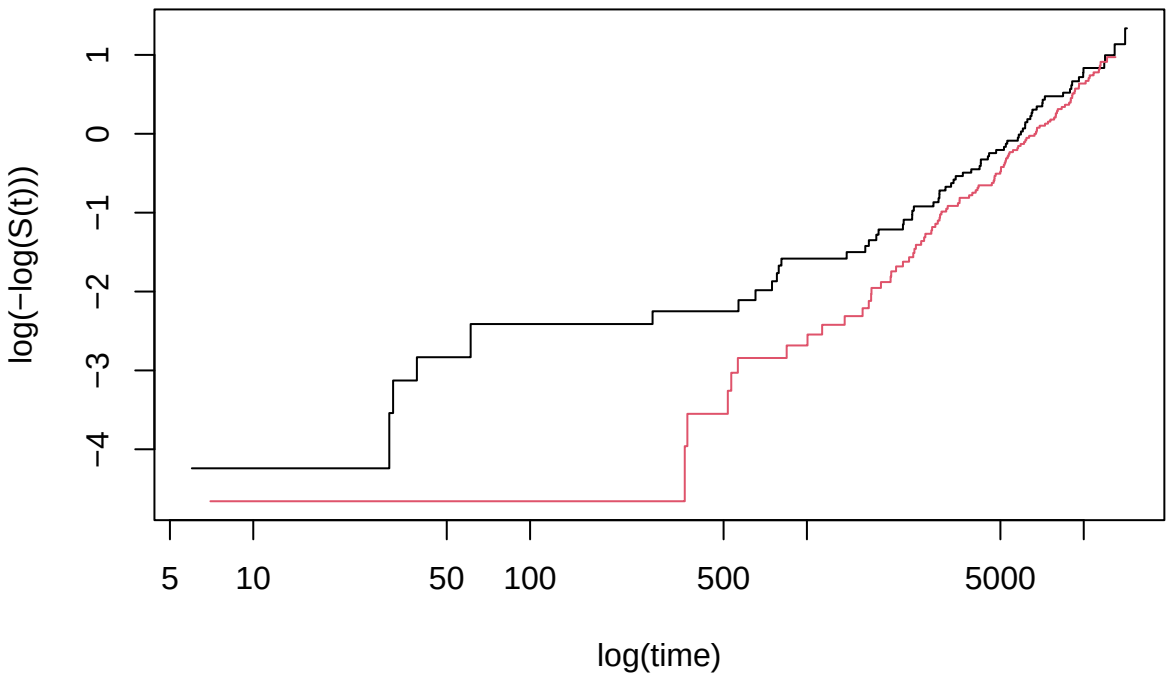
Maybe we check the semi-important predictors too (hgb, log_creat):

```r
# Partition age into categories
df$hgb_cat <- ifelse(df$hgb > median(df$hgb), "less", "more")
```

```r
surv_fit <- survfit(Surv(futime, death) ~ hgb_cat, data = df)

# Log-Log Plot
plot(surv_fit, fun = "cloglog", col = 1:2, xlab = "log(time)", ylab = "log(-log(S(t)))")
```




```r
# Partition age into categories
df$log_creat_cat <- ifelse(df$log_creat > median(df$log_creat), "less", "more")
surv_fit <- survfit(Surv(futime, death) ~ log_creat_cat, data = df)

# Log-Log Plot
plot(surv_fit, fun = "cloglog", col = 1:2, xlab = "log(time)", ylab = "log(-log(S(t)))")
```



Yes, very parallel for age. Only crosses at the very end where we have limited data for the other two variables. This supports the cox proportional odds assumption.

We interpret the model parameters by looking at the exp(coef) aka the hazard ratios. We see that the model predicts that every year increase in age increases the hazard of death by 7%, being male at 19% higher, for each unit increase of hbg a 11% decrease, for each unit increase of alb a 22% decrease, and finally for each unit of log_creat, hazard increases by over 70%.

Problem 3

The data set denim in the Faraway library contains the amount of waste in denim due to cutting the material for jeans. The data are broken down by five suppliers, which we'll assume are randomly sampled from a larger population of suppliers.

a.)

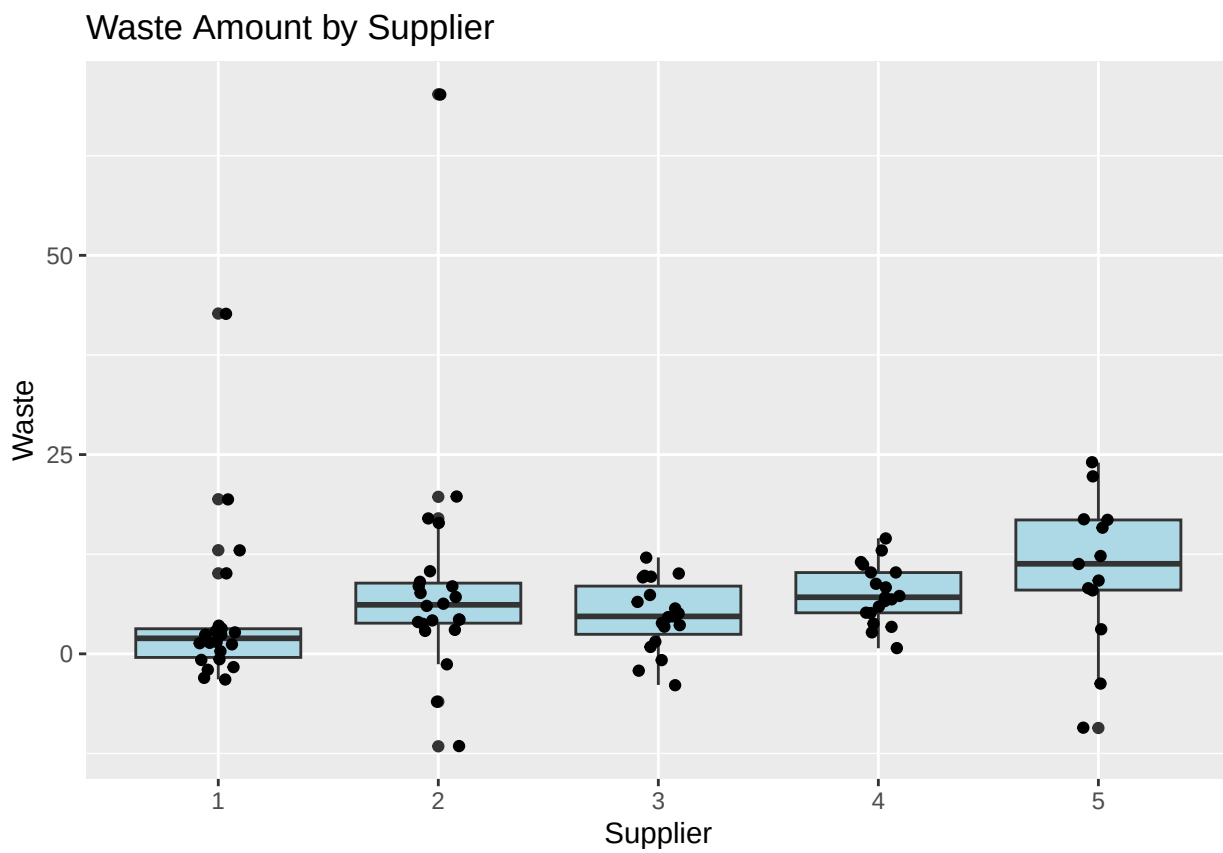
Visualize the data and comment on what you find.

Basic plot looking at the suppliers and waste

```r
library(faraway)
```

```
##
## Attaching package: 'faraway'

## The following objects are masked from 'package:survival':
##
##     rats, solder
```

```r
library(ggplot2)

data(denim)
ggplot(denim, aes(x = supplier, y = waste)) +
  geom_boxplot(fill = "lightblue") +
  geom_jitter(width = 0.1) +
  labs(title = "Waste Amount by Supplier", x = "Supplier", y = "Waste")
```



The variance between the suppliers definitely seems a bit different. The boxes are different heights suggesting some suppliers have more variation than others. Some also have huge outliers.

b.)

> Fit the linear mixed model and check assumptions. Can we trust the inference results? Explain your answer.

We use nlme and lme4 packages that Prof Song uses:

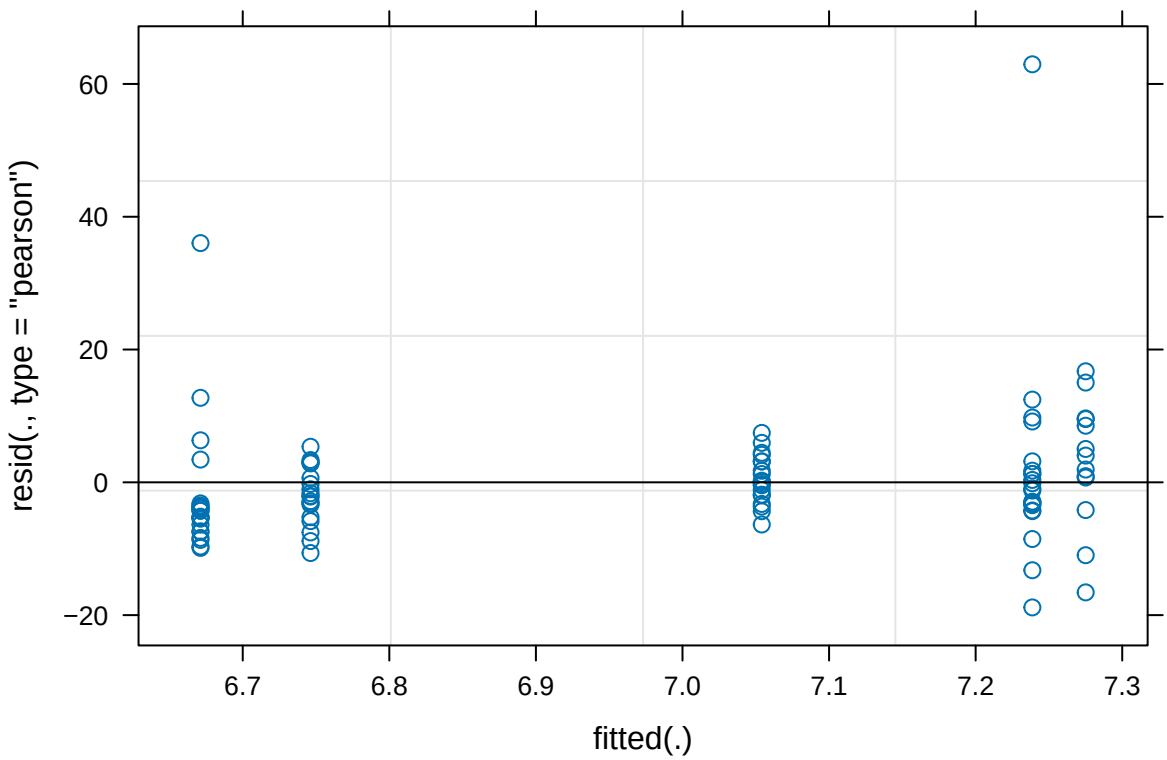
```
library(lme4)
```

```
## Loading required package: Matrix

##
## Attaching package: 'lme4'

## The following object is masked from 'package:faraway':
##
##     toenail
```

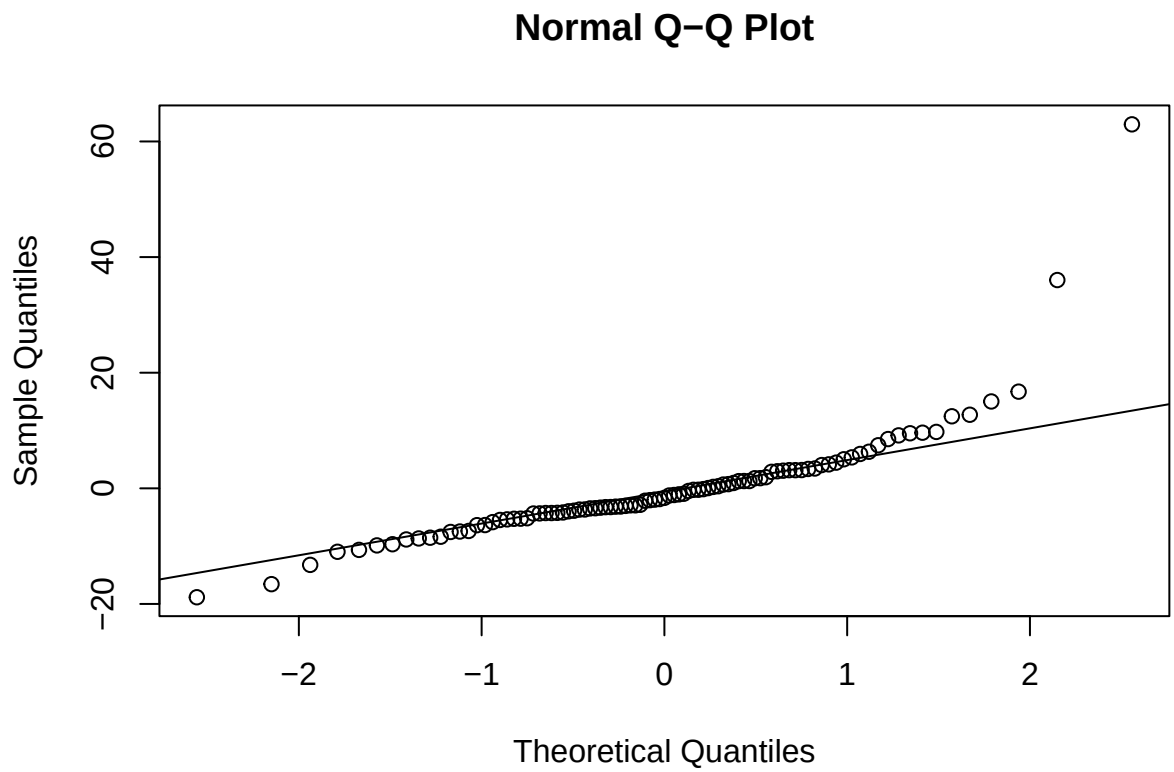
```
mix_model <- lmer(waste ~ 1 + (1|supplier), data = denim)
summary(mix_model)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: waste ~ 1 + (1 | supplier)
##    Data: denim
##
## REML criterion at convergence: 702.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.9095 -0.4363 -0.1669  0.3142  6.3817
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  supplier (Intercept)  0.6711  0.8192
##  Residual             97.3350  9.8658
## Number of obs: 95, groups:  supplier, 5
##
## Fixed effects:
##             Estimate Std. Error t value
## (Intercept)    6.997      1.078    6.49
```

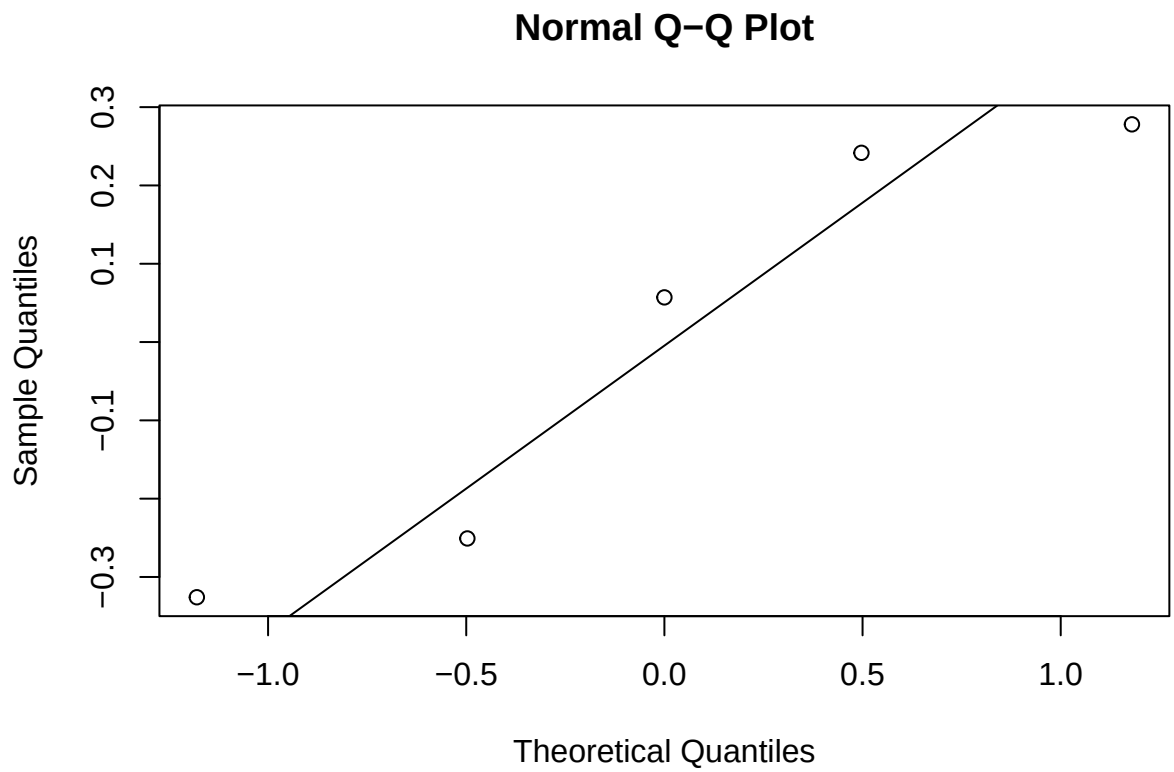
```r
plot(mix_model) # Pearson Residuals vs Fitted
```




```r
qqnorm(residuals(mix_model)); qqline(residuals(mix_model)) # Normality of errors
```




```
qqnorm(ranef(mix_model)$supplier[[1]]); qqline(ranef(mix_model)$supplier[[1]]) # Normality of random ef
```



We are using some diagnostic plots to understand whether the linear mixed model fits. The Pearson residuals do seem generally distributed around 0 line, with only a little bit of pattern. We are assuming linear models with normal error. This means that we want it to match QQ plot pretty closely. We see that this is generally the case for both error and for the randomness of supplier. However, it is a bit harder to diagnose the randomness of supplier due to the limited sample.

c.)

> Stats the null and alternative hypotheses of the ANOVA F test, the test statistic, degrees of freedom, and its P-value. What do you conclude?

For the one-way random effects model, our null hypothesis states that there is no variation between the suppliers, ie H_0: $\sigma_\alpha^2 = 0$. The alternative hypothesis would state that there is some variation, H_a: $\sigma_\alpha^2 > 0$.

The test statistic is F value.

```r
anova(lm(waste ~ supplier, data = denim))
```

```
## Analysis of Variance Table
##
## Response: waste
##           Df Sum Sq Mean Sq F value Pr(>F)
## supplier   4  450.9 112.730  1.1596  0.334
## Residuals 90 8749.1  97.212
```

Since p > 0.05, we do not reject H_0 and conclude that suppliers do not contribute significantly to total variance in waste.

d.)

Provide a 95% confidence interval for

```r
confint(mix_model, method = "profile")
```

```
## Computing profile confidence intervals ...

## Warning in nextpar(mat, cc, i, delta, lowcut, upcut): Last two rows have
## identical or NA .zeta values: using minstep
## Warning in nextpar(mat, cc, i, delta, lowcut, upcut): Last two rows have
## identical or NA .zeta values: using minstep

## Warning in FUN(X[[i]], ...): non-monotonic profile for .sig01

## Warning in confint.thpr(pp, level = level, zeta = zeta): bad spline fit for
## .sig01: falling back to linear interpolation

## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
## collapsing to unique 'x' values

##                2.5 %    97.5 %
## .sig01      0.000000  4.206374
## .sigma      8.591567 11.424996
## (Intercept) 4.977790  8.975895
```

1.) We are 95% confident that the true mean of waste is in the interval of [4.98, 8.96].

2.) We are 95% confident that the true variance of the error .sig01 (σ_α) between suppliers is [0, 4.21]. We are 95% confident that the true residual standard deviation .sigma (σ) is [8.59, 11.42].

3.) $\rho = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma^2}$

We can calculate the confidence interval because we assume MSTrt and MSE are independent chi-squared random variables. We get all we need from our anova table:

```r
anova_table <- anova(lm(waste ~ supplier, data = denim))

# Extract values
df1 <- anova_table$Df[1]           # Degrees of freedom for supplier
df2 <- anova_table$Df[2]           # Degrees of freedom for residuals
MSTrt <- anova_table$`Mean Sq`[1]  # MSTrt
MSE <- anova_table$`Mean Sq`[2]

# Get 'n' (observations per group)
# Since denim is balanced, we can just take the total N / number of groups
n <- nrow(denim) / (df1 + 1)

alpha <- 0.05
f_low <- qf(1 - alpha/2, df1, df2)
f_high <- qf(alpha/2, df1, df2)
```

```r
# Calculate L and U
L0 <- (1/n) * (MSTrt / (MSE * f_low) - 1)
U0 <- (1/n) * (MSTrt / (MSE * f_high) - 1)

# Only care about positive values
L <- max(0, L0)
U <- max(0, U0)

L / (L + 1)
```

```
## [1] 0
```

```r
U / (U + 1)
```

```
## [1] 0.3129748
```

We are 95% confident that the true intercorrelated correlation value lies between [0,0.313].

e.)

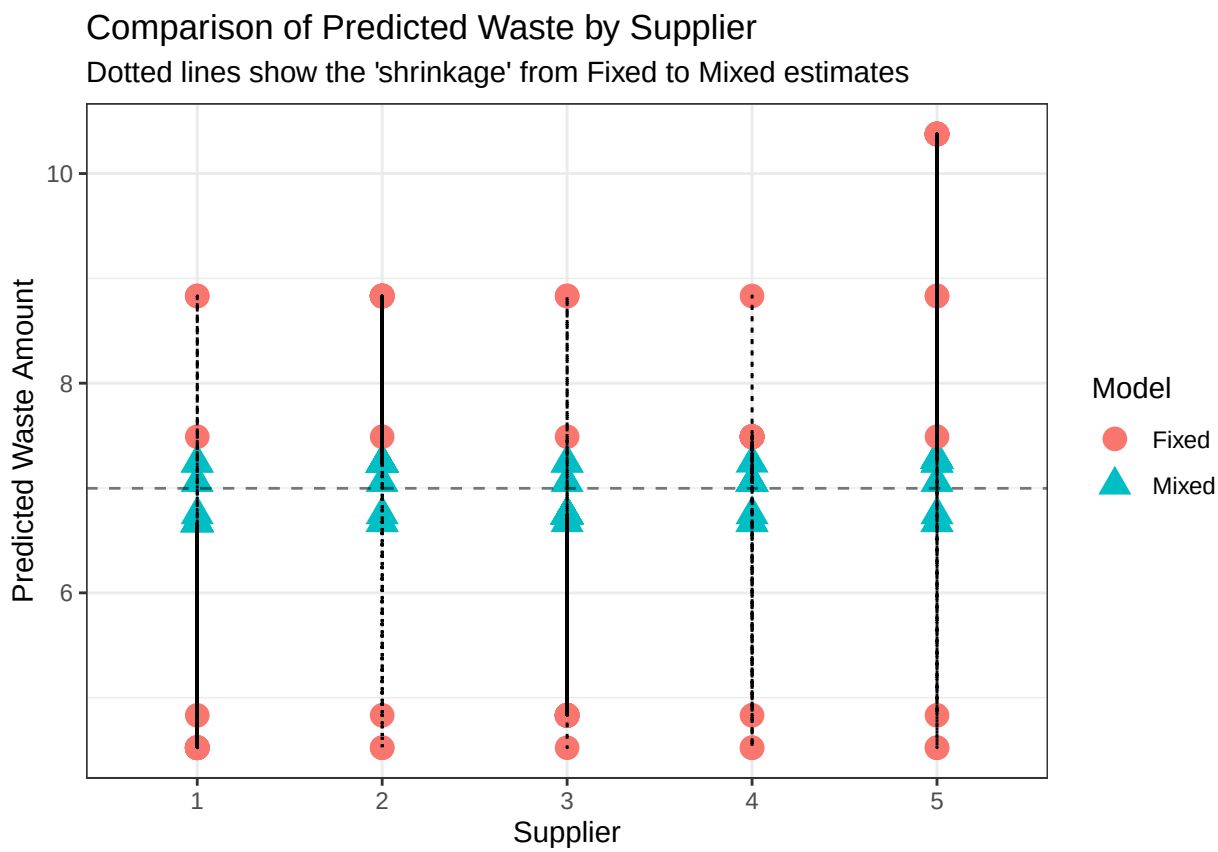
> Compare the fitted values $\hat{Y}_{ij}$ in the linear mixed model with the fitted values that would have been obtained if Supplier was viewed as fixed, and explain the reason for the difference in the estimates (if any).

```r
# Random effects predictions (mixed model)
ranef_fits <- predict(mix_model)
# Fixed effects predictions (regular lm)
fixef_fits <- predict(lm(waste ~ supplier, data = denim))

comp_data <- data.frame(
  supplier = levels(denim$supplier),
  Fixed = as.numeric(fixef_fits),
  Mixed = as.numeric(ranef_fits)
)

library(tidyr)
```

```
##
## Attaching package: 'tidyr'

## The following objects are masked from 'package:Matrix':
##
##     expand, pack, unpack
```

```r
plot_df <- pivot_longer(comp_data, cols = c("Fixed", "Mixed"),
                        names_to = "Model", values_to = "Predicted")

ggplot(plot_df, aes(x = supplier, y = Predicted, color = Model, shape = Model)) +
  geom_point(size = 4) +
  geom_line(aes(group = supplier), color = "black", linetype = "dotted") +
```

```r
geom_hline(yintercept = fixef(mix_model), linetype = "dashed", alpha = 0.5) +
labs(title = "Comparison of Predicted Waste by Supplier",
     subtitle = "Dotted lines show the 'shrinkage' from Fixed to Mixed estimates",
     x = "Supplier",
     y = "Predicted Waste Amount") +
theme_bw()
```



Pretty cool that we can see that the mixed model has predicted values much closer to the global mean. This is probably because the mixed model sees the supplier as extra variation / error and controls for that while the fixed model does not.

Problem 4

A school superintendent is concerned about the development of technology skills for research in high school. Since there are four high schools in his district, all of which go about this instruction differently, he decided to assess if there were any differences across schools. He first compiled a long list of "tech skills" and randomly selected four to be used in his study. He then randomly selected 24 students from each school and assigned each to one of the four tasks so that there were six students per task per school. Each student then performed the skill and was scored using a 0-100 numeric scale.

a.)

If a two-way ANOVA is to be used for the analysis, should it be treated as a fixed effects, random effects, or mixed effects model? State the model you would use and its assumptions.

This is a mixed effects model. School is the fixed effect. The skill that is being tested is the random effect because they are chosen randomly out of a long list.

My model is:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

μ is the grand mean, α_i is the fixed effect of the i-th School, β_j is the random effect of the j-th Skill with $\beta_j \sim N(0, \sigma^2_\beta)$, $(\alpha\beta)_{ij}$ is the random interaction effect with $(\alpha\beta)_{ij} \sim N(0, \sigma^2_{\alpha\beta})$, ϵ_{ijk} is the random error with $\epsilon_{ijk} \sim N(0, \sigma^2)$.

Since it is a linear mixed model, we assume independence between observation, that the error terms and random effects are normally distributed, and finally that the variance is constant across groups.

b.)

Complete the following ANOVA table and determine which effects are significant at the $\alpha = .05$ level. State your conclusions, making sure to estimate all variances and describing any additional mean comparisons you'd like to perform.

Source	DF	SS	MS	F-statistic
School	3	222.0	74	13.15
Skill	3	96.0	32	5.68
School × Skill	9	135.0	15	2.66
Error	80	450.0	5.625	-
Total	95	903.0	-	-

We compare these F statistics against their F values. School, skill, school x skill are all significant. Our variance estimates are $\hat{\sigma}^2 = MS_E = \mathbf{5.625}$, $\hat{\sigma}^2_{\alpha\beta} = \frac{MS_{AB} - MS_E}{n} = \frac{15-5.625}{6} = \mathbf{1.5625}$, $\hat{\sigma}^2_\beta = \frac{MS_B - MS_E}{an} = \frac{32-5.625}{24} \approx \mathbf{1.10}$.

Since we know school effect is significant, we can perform Tukey's tests to compare the specific schools with one another.

c.)

> If the grand skill level of the high schools (average over the four schools) is of interest, describe how one would construct a 95% confidence interval.

Our point estimate for the grand skill is the mean of the whole sample set. We then compute standard error by estimating the variance (assuming independent normal errors) $\text{Var}(\bar{Y}_{...}) = \frac{\sigma_\beta^2}{b} + \frac{\sigma_{\alpha\beta}^2}{ab} + \frac{\sigma^2}{abn}$ where $a = 4$, $b = 4$, and $n = 6$. We would plug in our estimated variances. We take the square root to get SE. We have a t distribution, where we have significant error from skill. We could use its degrees of freedom, so t($\alpha/2$, df=3).

Finally

$$\bar{Y}_{...} \pm t_{\alpha/2, df=3} \cdot SE$$